Purple Comet! Math Meet 2026

High School Division

Problems, Solutions and Remarks

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HU Collection

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The thirty problems below are the official High School problems of the April 2026 Purple Comet! Math Meet, © Titu Andreescu and Jonathan Kane. The solutions, remarks and extension challenges are my own. Every answer has been checked independently, but I would be grateful to hear of any slips through the website above.

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Every entry above is a hyperlink. Each problem statement also carries a link to its solution, and each solution links back to the statement.

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A note on the collection

Purple Comet rewards a particular kind of speed: not the speed of a calculator, but the speed of recognising which of five or six standard ideas a problem is built on. Almost nothing here needs more than two such ideas. The solutions are written to make those ideas crystal clear, with computation kept to what is needed to certify the answer.

Where a problem shows significant potential (i.e. a Fibonacci count in a random walk, a telescoping sum in a Sophie Germain factorisation, a chromatic polynomial in a colouring count), I have added a remark or an extension challenge. The extensions are meant to be worked, not read; answers are given, but proofs aren't, and it is strongly advised that the reader attemps them first.

Answers are collected in the key at the end. Diagrams are drawn to scale with `tkz-euclide`.

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1 Problems

Problem 1

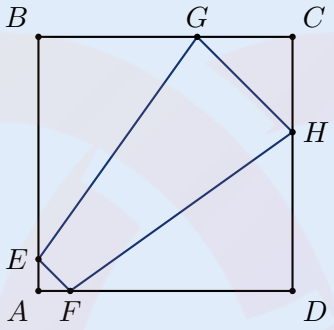
One afternoon at a furniture store, Heather bought a 100 dollar chair at a 20 percent discount and a 150 dollar table at a 30 percent discount. Find the percent discount she received on her afternoon's purchases. [Solution 1]

Problem 2

Find the number of prime numbers $p > 0$ such that at least 50 positive integers less than 2026 are divisible by p. [Solution 2]

Problem 3

Square $ABCD$ has side length 8. Points E and F are on AB and AD, respectively, with $AE = AF = 1$, and points G and H are on BC and CD, respectively, with $CG = CH = 3$, as shown. Find the area of trapezoid $EGHF$. [Solution 3]



Problem 4

The mean of a list of 21 numbers is 21. The mean of the first 17 numbers in the list is 17. Find the mean of the last 4 numbers in the list. [Solution 4]

Problem 5

Find the least positive integer n such that n, $n+1$, $n+2$, and $n+3$ are each divisible by exactly two different prime numbers. [Solution 5]

Problem 6

For a given positive integer n, let $A = 1 + 2 + 3 + \cdots + n$ and $B = 1^2 + 2^2 + 3^2 + \cdots + n^2$. Find the value of n such that $\log_3 A$ and $\log_3 B$ differ by 5. [Solution 6]

Problem 7

One way to write 2026 as the sum of three perfect squares is $1^2 + 27^2 + 36^2$. Find the greatest possible value of s, where q, r, and s are positive integers with $q^2 + r^2 + s^2 = 2026$. [Solution 7]

Problem 8

A train traveling at a constant speed takes 15 seconds to pass a point on its track, and 45 seconds to pass completely through a tunnel that is 600 meters long. Find the speed of the train in kilometers per hour. [Solution 8]

Problem 9

You have three pieces of red paper, three pieces of white paper, and three pieces of blue paper. You have three black stickers, three yellow stickers, and three purple stickers. You randomly place one sticker onto each piece of paper. The probability that you will end up with all nine

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possible color combinations is $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m+n$. [Solution 9]

Problem 10

Jacob had 25% more in savings than his son Sawyer, and when they pooled their savings, they were 25% short of having the money they needed to purchase the used car they wanted. After working hard for a week, Jacob saved an additional amount, and Sawyer saved an additional amount that was 25% more than the additional amount Jacob saved. Now when they pooled their savings, they had just enough money to purchase the car. The portion of the purchase price of the car that Jacob contributed was $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m+n$. [Solution 10]

Problem 11

Let $A = \{1, 2, 3, \ldots, 2026\}$. Let N be the number of ways that the 2026 elements of A can be paired into 1013 sets of 2 elements each such as

$$\{1, 2\}, \{3, 4\}, \{5, 6\}, \ldots, \{2025, 2026\}.$$

Find the greatest integer n such that 63^n divides N. [Solution 11]

Problem 12

Find the number of positive integers that divide neither 2026^2 nor 6000^2 but do divide the product $2026^2 \cdot 6000^2$. [Solution 12]

Problem 13

One marker is placed at each of the points 1, 2, 3, and 4 along a horizontal number line. Then a shift takes place where each marker either moves one unit to the left, stays where it began, or moves one unit to the right, where the choices for each marker are made randomly and independently of the choices for the other markers. The probability that after the shift there is at least one location with two or more markers is $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m+n$. [Solution 13]

Problem 14

Circles with radii 6 and 18 are externally tangent. The region bounded by the two circles and one of their common external tangents has area $m\sqrt{p} - n\pi$, where m, n, and p are positive integers, and p is not divisible by the square of any prime. Find $m+n+p$. [Solution 14]

Problem 15

Let a be a real number for which $20 \sin a + 21 \cos a = 29$. Then $20 \sin(3a) + 21 \cos(3a) = \frac{m}{n}$, where m and n are relatively prime positive integers. Find $m+n$. [Solution 15]

Problem 16

The sum of the lengths of the diagonals of a rhombus is $\frac{17}{26}$ times the perimeter of the rhombus. The cosine of the acute internal angles of the rhombus is $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m+n$. [Solution 16]

Problem 17

A swimming pool can be filled using any of three faucets A, B, and C, which when opened individually can fill the pool in 2, 3, and 6 hours, respectively. Suppose that starting when the pool is empty, faucets A and B are opened simultaneously, and after k minutes, when the pool is partially filled, faucet C is opened resulting in the pool being filled in a total of 1

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hour and 3 minutes. Find k. [Solution 17]

Problem 18

In the complex plane, the triangle with vertices at 0, z, and z^2 has centroid $17+69i$, where $i=\sqrt{-1}$. Find the least possible value of $|z|^2$. [Solution 18]

Problem 19

Let $p(x)$ be the polynomial with minimum degree, with integer coefficients, and with leading coefficient equal to 1 such that $\sqrt{\sqrt{3}-\sqrt{2}}$ is a root of p. Find $p(2)$. [Solution 19]

Problem 20

A point is randomly selected from the inside of a triangle with side lengths 17, 19, and 22. The probability that the point is closer to the longest side of the triangle than to either of the other two sides is $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m+n$. [Solution 20]

Problem 21

Bob has two identical-looking coins, but one is a fair coin that is equally likely to come up heads or tails when it is flipped, and the other is an unbalanced coin that comes up heads $\frac{3}{5}$ of the time when it is flipped. Suppose Bob randomly selects one of the coins, flips it, and it comes up heads. The probability that the same coin will come up heads when Bob flips it a second time is $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m+n$. [Solution 21]

Problem 22

Each cell of a 2×4 grid is colored either red, white, blue, or yellow. Find the number of ways the cells can be colored so that no two cells that share a side have the same color, and each of the four colors is used on at least one cell, such as in the example shown below. [Solution 22]



Problem 23

The equation $x^4-56x+15=0$ has a complex solution of the form $-a+i\sqrt{b}$, where a and b are positive integers and $i=\sqrt{-1}$. Find $a+10b$. [Solution 23]

Problem 24

Four mutually externally tangent spheres with centers A, B, C, and D have radii 1, 2, 3, and 4, respectively. There are relatively prime positive integers m and n such that the volume of the tetrahedron (triangular pyramid) $ABCD$ is $\sqrt{\frac{m}{n}}$. Find $m+n$. [Solution 24]

Problem 25

Four red, four white, and four blue marbles are randomly distributed to Alan, Beth, Carl, and Dora, so that each person receives three marbles. The probability that Alan does not receive any red marbles and Dora does not receive any blue marbles is $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m+n$. [Solution 25]

Problem 26

Let x and y be the positive real numbers satisfying $xy=16$ for which the value of $3x+y^3$ is a minimum. Let m be that minimum value. Find $x+y+m$. [Solution 26]

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Problem 27

A right circular cone has height 12 and base diameter 10. A hemisphere is internally tangent to the side of the cone, and the base of the hemisphere lies in the base of the cone so that the hemisphere and the base of the cone have the same center. The largest sphere internally tangent to the side of the cone and externally tangent to the hemisphere has radius $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m+n$. [Solution 27]

Problem 28

For positive integers n, define

$$a_n = 1 - \frac{2n^2}{1+\sqrt{1+4n^4}}.$$

Evaluate $\sqrt{a_1} + 2\sqrt{a_2} + 3\sqrt{a_3} + \cdots + 119\sqrt{a_{119}}$. [Solution 28]

Problem 29

You have six red bricks, six white bricks, and six blue bricks. You are to put down the bricks in three side-by-side piles of six bricks each so that stacks of red bricks are only at the bottoms of piles, and stacks of blue bricks are only at the tops of piles. Four possible patterns are shown below. Find the number of possible color patterns you can get. [Solution 29]



(Illustrative valid patterns; the official figure may differ.)

Problem 30

Inside a sphere with radius 9 is a cube with edge length 8. The four vertices of the bottom face of the cube rest against the sphere. Two smaller congruent spheres are externally tangent to each other, are both externally tangent to the top face of the cube, and are internally tangent to the larger sphere. The greatest possible radius of the smaller spheres can be written as $\sqrt{m}-n$, where m and n are positive integers. Find $m+n$. [Solution 30]

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2 Solutions

2.1 Problem 1

[Statement]

Heather pays $80 + 105 = 185$ dollars for goods priced at 250, so the discount is $65/250 = 26\%$. Note that the combined discount is a price-weighted average of 20% and 30%, weighted $100 : 150$, which is another way to see 26.

Answer. 26

2.2 Problem 2

[Statement]

The positive integers below 2026 divisible by p are $p, 2p, \ldots$, and there are $\lfloor 2025/p \rfloor$ of them. This is at least 50 exactly when $p \le 40$, since $2025/50 = 40.5$. The primes up to 40 are $2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37$, twelve in all. (For $p = 41$ the count is 49, so the cut is sharp.)

Answer. 12

2.3 Problem 3

[Statement]

Both EF and GH are perpendicular to the diagonal AC, which is why $EGHF$ is a trapezoid. Measured along AC, the line EF sits at distance $\frac{1}{\sqrt{2}}$ from A and the line GH at distance $\frac{3}{\sqrt{2}}$ from C, so the height of the trapezoid is $8\sqrt{2} - \frac{4}{\sqrt{2}} = 6\sqrt{2}$. The parallel sides have lengths $EF = \sqrt{2}$ and $GH = 3\sqrt{2}$. Hence

$$[EGHF] = \tfrac{1}{2}\big(\sqrt{2} + 3\sqrt{2}\big) \cdot 6\sqrt{2} = 24.$$

As a check, subtracting the four corner triangles from the square gives $64 - \frac{1}{2} - \frac{9}{2} - \frac{35}{2} - \frac{35}{2} = 24$.

Answer. 24

2.4 Problem 4

[Statement]

The whole list sums to $21 \cdot 21 = 441$ and the first 17 numbers sum to $17 \cdot 17 = 289$, so the last four sum to 152 and have mean 38.

Answer. 38

2.5 Problem 5

[Statement]

List the integers with exactly two distinct prime factors: $6, 10, 12, 14, 15, 18, 20, 21, 22, 24, 26, 28, 33, 34, 35, 36,$ The first run of four consecutive terms is

$$33 = 3 \cdot 11, \quad 34 = 2 \cdot 17, \quad 35 = 5 \cdot 7, \quad 36 = 2^2 \cdot 3^2.$$

Below this the longest run is $20, 21, 22$, broken by the prime 23.

Answer. 33

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Extension challenge

Find the least n for which five consecutive integers $n, \ldots, n+4$ each have exactly two distinct prime factors, and then the least n for six consecutive integers. (Answers: 54 and 91. The run starting at 91 has length exactly six, since 97 is prime. The least run of length seven starts at 141 and turns out to have length eight. Note that any run of length six or more contains a multiple of 6, which must be of the form $2^a 3^b$; such numbers are sparse, which limits how long runs can be.)

2.6 Problem 6

[Statement]

With $A = \frac{n(n+1)}{2}$ and $B = \frac{n(n+1)(2n+1)}{6}$ we have $B/A = \frac{2n+1}{3}$. The logarithms differ by 5 when $B/A = 3^5 = 243$, so $2n+1 = 729$ and $n = 364$.

Answer. 364

2.7 Problem 7

[Statement]

Since $45^2 = 2025$ leaves $q^2 + r^2 = 1$, impossible for positive q, r, we need $s \le 44$. Now $2026 - 44^2 = 90 = 9^2 + 3^2$, so $s = 44$ is attained: $2026 = 3^2 + 9^2 + 44^2$.

Answer. 44

2.8 Problem 8

[Statement]

Let the train have length L and speed v in m/s. Passing a point takes $L/v = 15$ seconds; clearing the tunnel takes $(L+600)/v = 45$ seconds. Subtracting, $600/v = 30$, so $v = 20$ m/s $= 72$ km/h.

Answer. 72

2.9 Problem 9

[Statement]

Only the colour pattern matters. There are $\binom{9}{3,3,3} = 1680$ ways to distribute the three sticker colours among the nine papers. To realise all nine combinations, each colour class of paper must receive one sticker of each colour, which can happen in 3! ways per paper colour, $6^3 = 216$ ways in all. The probability is $216/1680 = 9/70$.

Answer. 79

2.10 Problem 10

[Statement]

Let Sawyer start with s; Jacob starts with $\frac{5}{4}s$. Their pool $\frac{9}{4}s$ is 75% of the price P, so $P = 3s$. If Jacob then adds x and Sawyer adds $\frac{5}{4}x$, the shortfall $\frac{3}{4}P = \frac{9}{4}s$ equals $\frac{9}{4}x$, so $x = s$. Jacob's total is $\frac{5}{4}s + s = \frac{9}{4}s$ out of $3s$, that is $\frac{3}{4}$ of the price.

Answer. 7

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2.11 Problem 11

[Statement]

The number of perfect matchings of a $2m$-set is $(2m-1)!! = 1\cdot 3\cdot 5\cdots(2m-1)$, so $N = 2025!!$. Since $63 = 3^2\cdot 7$ we need $\nu_3(N)$ and $\nu_7(N)$. For odd p, the odd multiples of p^k up to 2025 number $\lfloor 2025/p^k\rfloor - \lfloor 2025/2p^k\rfloor$, so

$$\nu_3(N) = (675-337)+(225-112)+(75-37)+(25-12)+(8-4)+(2-1) = 507,$$
$$\nu_7(N) = (289-144)+(41-20)+(5-2) = 169.$$

Thus $\nu_{63}(N) = \min\big(\lfloor 507/2\rfloor, 169\big) = 169$.

Answer. $\boxed{169}$

Remark

Equivalently $N = \dfrac{2026!}{2^{1013}\,1013!}$, so for odd p one has $\nu_p\big((2m-1)!!\big) = \nu_p\big((2m)!\big) - \nu_p(m!)$, a Legendre-type formula for double factorials. The subtraction form above is the same computation term by term.

2.12 Problem 12

[Statement]

Factor: $2026 = 2\cdot 1013$ with 1013 prime, and $6000 = 2^4\cdot 3\cdot 5^3$. Hence

$$2026^2 = 2^2\cdot 1013^2, \qquad 6000^2 = 2^8\cdot 3^2\cdot 5^6, \qquad 2026^2\cdot 6000^2 = 2^{10}\cdot 3^2\cdot 5^6\cdot 1013^2.$$

The product has $11\cdot 3\cdot 7\cdot 3 = 693$ divisors; 2026^2 has 9 and 6000^2 has 189, and they share the 3 divisors of gcd $= 2^2$. By inclusion-exclusion the answer is $693-(9+189-3) = 498$.

Answer. $\boxed{498}$

2.13 Problem 13

[Statement]

We count the complementary event: the final positions $x_i = i + d_i$ with $d_i \in \{-1,0,1\}$ are all distinct. Condition on marker 1.

- $x_1 = 0$. Markers 2, 3, 4 are then unconstrained by marker 1; this is the same problem for three markers, which one checks has 14 good outcomes (27 total, 13 with a collision).
- $x_1 = 1$. Marker 2 must go to 2 or 3. If $x_2 = 2$, markers 3, 4 avoid each other in $3\cdot 2 - 2 = 4$ ways; if $x_2 = 3$, then $x_3 \in \{2,4\}$, $x_4 \in \{4,5\}$ distinct gives 3 ways. Total 7.
- $x_1 = 2$. Marker 2 must go to 1 or 3. If $x_2 = 1$, markers 3, 4 give 4 ways as before; if $x_2 = 3$, marker 3 is forced to 4 and marker 4 to 5. Total 5.

Altogether $14+7+5 = 26$ collision-free outcomes among $3^4 = 81$, so the probability of a collision is $1 - \frac{26}{81} = \frac{55}{81}$.

Answer. $\boxed{136}$

Remark

The casework has a clean structure. Let $h(n)$ be the number of collision-free shifts of n markers and $g(n)$ the number when the first marker is forbidden to move left. Conditioning on the first one or two markers gives

$$g(n) = g(n-1)+g(n-2)+1, \qquad h(n) = h(n-1)+g(n-1)+g(n-2)+1,$$

with $g(0) = 1$, $g(1) = 2$, $h(0) = 1$. So $g(n) = F_{n+3} - 1$ and, summing, $h(n) = F_{n+5} - n - 4$,

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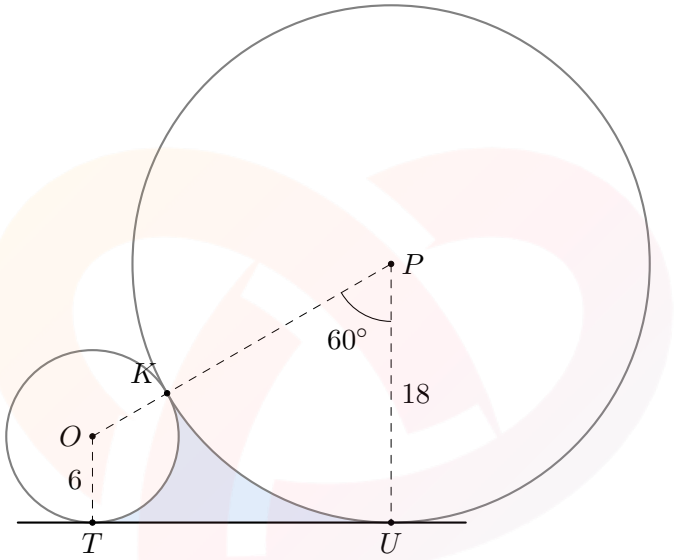
where F_k are the Fibonacci numbers. Indeed $h(4) = F_9 - 8 = 34 - 8 = 26$.

Extension challenge

Prove the formula $h(n) = F_{n+5} - n - 4$ above, and deduce that for n markers the probability of a collision is $1 - \dfrac{F_{n+5} - n - 4}{3^n}$. Then explain combinatorially why $g(n) + 1$ is a Fibonacci number: which familiar Fibonacci-counted objects do the collision-free shifts with no left move correspond to? (For $n = 5$ the count $h(5)$ is $55 - 9 = 46$; for $n = 6$ it is $89 - 10 = 79$.)

2.14 Problem 14

[Statement]



Let the circles have centres O (radius 6) and P (radius 18), touching at K, with tangent points T and U on the common tangent. Then $OP = 24$ and $TU = \sqrt{24^2 - 12^2} = 12\sqrt{3}$, so the right trapezoid $OTUP$ has area $\frac{1}{2}(6 + 18) \cdot 12\sqrt{3} = 144\sqrt{3}$.

In this trapezoid, $\cos \angle OPU = \frac{18-6}{24} = \frac{1}{2}$, so $\angle OPU = 60°$ and $\angle POT = 120°$. The shaded region is the trapezoid minus a $60°$ sector of the large circle and a $120°$ sector of the small one:

$$144\sqrt{3} - \tfrac{60}{360}\pi \cdot 18^2 - \tfrac{120}{360}\pi \cdot 6^2 = 144\sqrt{3} - 54\pi - 12\pi = 144\sqrt{3} - 66\pi.$$

So $m + n + p = 144 + 66 + 3 = 213$.

Answer. $\boxed{213}$

Extension challenge

For externally tangent circles of radii $r < R$, show that the corresponding region has area

$$(R + r)\sqrt{Rr} - \tfrac{1}{2}\big(R^2\theta + r^2(\pi - \theta)\big), \qquad \cos\theta = \frac{R - r}{R + r}.$$

For which ratios R/r is θ a rational multiple of π with a rational cosine? (Only $R/r \in \{1, 3\}$ give $\theta \in \{90°, 60°\}$; the problem's $18 : 6$ is the nice case.)

2.15 Problem 15

[Statement]

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Since $20^2 + 21^2 = 29^2$, write $20\sin a + 21\cos a = 29\sin(a+\varphi)$ with $\cos\varphi = \frac{20}{29}$, $\sin\varphi = \frac{21}{29}$. Equality with 29 forces $a + \varphi \equiv \frac{\pi}{2}$, so $\sin a = \cos\varphi = \frac{20}{29}$ and $\cos a = \sin\varphi = \frac{21}{29}$. Then

$$20\sin 3a + 21\cos 3a = 29\big(\sin 3a\sin a + \cos 3a\cos a\big) = 29\cos 2a = 29\cdot\frac{21^2 - 20^2}{29^2} = \frac{41}{29}.$$

Answer. $\boxed{70}$

Remark

The same argument shows that if $A\sin a + B\cos a = \sqrt{A^2+B^2}$, then for every integer k,

$$A\sin(ka) + B\cos(ka) = \sqrt{A^2+B^2}\,\cos\big((k-1)a\big).$$

The case $k = 3$ is the problem; $k = 2$ gives $29\cos a = 21$, which is a useful sanity check.

2.16 Problem 16

[Statement]

Let the side be s and an acute angle be θ. The diagonals are $2s\cos\frac{\theta}{2}$ and $2s\sin\frac{\theta}{2}$, so the condition reads $\cos\frac{\theta}{2} + \sin\frac{\theta}{2} = \frac{17}{13}$. Squaring, $1 + \sin\theta = \frac{289}{169}$, so $\sin\theta = \frac{120}{169}$ and, θ being acute, $\cos\theta = \frac{119}{169}$.

Answer. $\boxed{288}$

2.17 Problem 17

[Statement]

In pools per minute, $A + B$ fills $\frac{1}{120} + \frac{1}{180} = \frac{1}{72}$ and $A + B + C$ fills $\frac{1}{60}$. Therefore

$$\frac{k}{72} + \frac{63-k}{60} = 1 \implies 5k + 6(63-k) = 360 \implies k = 18.$$

Answer. $\boxed{18}$

2.18 Problem 18

[Statement]

The centroid condition is $z + z^2 = 51 + 207i$. Completing the square, $\big(z+\frac{1}{2}\big)^2 = \frac{205}{4} + 207i$. Put $w = z + \frac{1}{2} = x + yi$. Then $x^2 - y^2 = \frac{205}{4}$ and $x^2 + y^2 = |w^2| = \sqrt{\frac{205^2}{16} + 207^2} = \frac{853}{4}$, giving $x^2 = \frac{529}{4}$, $y^2 = 81$, so $w = \pm\big(\frac{23}{2} + 9i\big)$ and $z = 11 + 9i$ or $z = -12 - 9i$. The two values of $|z|^2$ are 202 and 225.

Answer. $\boxed{202}$

2.19 Problem 19

[Statement]

Let $x = \sqrt{\sqrt{3} - \sqrt{2}}$. Then $x^2 = \sqrt{3} - \sqrt{2}$, $x^4 = 5 - 2\sqrt{6}$, and $(x^4 - 5)^2 = 24$, so x is a root of $x^8 - 10x^4 + 1$. This is the minimal polynomial: $\sqrt{3} - \sqrt{2}$ has degree 4 over $\mathbb{Q}$, and it is not a square in $\mathbb{Q}(\sqrt{2}, \sqrt{3})$, because the square of $a + b\sqrt{2} + c\sqrt{3} + d\sqrt{6}$ has rational part $a^2 + 2b^2 + 3c^2 + 6d^2$, which vanishes only when $a = b = c = d = 0$. So x has degree 8 and $p(x) = x^8 - 10x^4 + 1$, whence $p(2) = 256 - 160 + 1 = 97$.

Answer. $\boxed{97}$

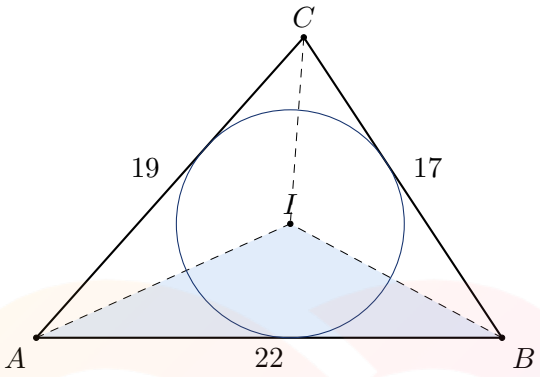
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Remark

The eight roots of p are $\pm\sqrt{\sqrt{3}\pm\sqrt{2}}$ and $\pm i\sqrt{\sqrt{3}\pm\sqrt{2}}$. Note also that $p(x) = (x^4-1)^2-8x^4$, which shows at a glance that p has no rational roots and hints at the factorisation over $\mathbb{Q}(\sqrt{2})$: $p(x) = (x^4 - 2\sqrt{2}x^2 - 1)(x^4 + 2\sqrt{2}x^2 - 1)$.

2.20 Problem 20

[Statement]



Label the triangle ABC with $AB = 22$, $BC = 17$, $CA = 19$ and incentre I. The points equidistant from sides AB and BC lie on the bisector from B, and those equidistant from AB and CA lie on the bisector from A; so the points closer to AB than to either other side form exactly the triangle ABI. Its area is $\frac{1}{2}\cdot 22\cdot r$, while $[ABC] = rs$ with $s = 29$. The probability is therefore

$$\frac{\frac{1}{2}\cdot 22\,r}{29\,r} = \frac{11}{29}.$$

Answer. 40

Remark

In any triangle the probability of being closest to the side of length a is $\dfrac{a}{a+b+c}$; the inradius cancels and the side lengths alone decide. The specific values 17 and 19 are decoration.

2.21 Problem 21

[Statement]

Each coin is chosen with probability $\frac{1}{2}$. The first flip is heads with probability $\frac{1}{2}\cdot\frac{1}{2}+\frac{1}{2}\cdot\frac{3}{5} = \frac{11}{20}$, so by Bayes the posterior odds are fair : biased $= 5 : 6$. Hence

$$P(\text{second heads} \mid \text{first heads}) = \frac{5}{11}\cdot\frac{1}{2} + \frac{6}{11}\cdot\frac{3}{5} = \frac{25+36}{110} = \frac{61}{110}.$$

Note $\frac{61}{110} > \frac{11}{20}$: seeing heads makes heads more likely, as it should.

Answer. 171

2.22 Problem 22

[Statement]

First count proper colourings of the 2×4 grid with k colours available. The first column can be coloured in $k(k-1)$ ways. Given a coloured column (a, b), the next column (c, d) needs $c \neq a$,

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$d \neq b$, $c \neq d$: if $c = b$ there are $k-1$ choices for d, and for each of the $k-2$ other choices of c there are $k-2$ choices for d. So each new column multiplies the count by $(k-1)+(k-2)^2 = k^2-3k+3$, and

$$P(k) = k(k-1)(k^2-3k+3)^3, \qquad P(4) = 4116,\ P(3) = 162,\ P(2) = 2,\ P(1) = 0.$$

By inclusion-exclusion over the set of colours actually used, the number of proper colourings using all four colours is

$$P(4) - \binom{4}{1}P(3) + \binom{4}{2}P(2) - \binom{4}{3}P(1) = 4116 - 648 + 12 - 0 = 3480.$$

Answer. $\boxed{3480}$

Extension challenge

$P(k)$ above is the chromatic polynomial of the ladder graph $P_2 \times P_4$. Show that for the $2 \times n$ grid the number of proper colourings using all four of four colours is $12 \cdot 7^{n-1} - 24 \cdot 3^{n-1} + 12$, and find the analogous transfer factor for the $3 \times n$ grid with k colours (it is no longer a single number; a 2-state transfer matrix appears).

2.23 Problem 23

[Statement]

With no cubic term, try $x^4 - 56x + 15 = (x^2 + px + q)(x^2 - px + r)$, which requires $q + r = p^2$, $p(r-q) = -56$, $qr = 15$. The choice $p = 4$, $q = 15$, $r = 1$ works:

$$x^4 - 56x + 15 = (x^2 + 4x + 15)(x^2 - 4x + 1).$$

The first factor has roots $-2 \pm i\sqrt{11}$, so $a = 2$, $b = 11$ and $a + 10b = 112$.

Answer. $\boxed{112}$

2.24 Problem 24

[Statement]

The edge lengths are sums of radii: $AB = 3$, $AC = 4$, $AD = 5$, $BC = 5$, $BD = 6$, $CD = 7$. Since $AB^2 + AC^2 = BC^2$, the angle at A in face ABC is right; place $A = (0,0,0)$, $B = (3,0,0)$, $C = (0,4,0)$. Solving $|D|^2 = 25$, $|D-B|^2 = 36$, $|D-C|^2 = 49$ gives $D = \left(-\frac{1}{3}, -1, \frac{\sqrt{215}}{3}\right)$. Hence

$$V = \frac{1}{3} \cdot [ABC] \cdot \frac{\sqrt{215}}{3} = \frac{1}{3} \cdot 6 \cdot \frac{\sqrt{215}}{3} = \frac{2\sqrt{215}}{3} = \sqrt{\frac{860}{9}}.$$

Answer. $\boxed{869}$

Remark

The right angle at A is what makes the coordinates clean: the pairwise sums $1+2$, $1+3$, $2+3$ form a 3-4-5 triple. With radii $1, 2, 3, \rho$ the base ABC is unchanged and the apex has $x = \frac{3-\rho}{3}$, $y = \frac{2-\rho}{2}$, so the volume is $2\sqrt{(1+\rho)^2 - x^2 - y^2}$. This vanishes exactly when $23\rho^2 + 132\rho - 36 = 0$, that is $\rho = \frac{6}{23}$, which is the radius of the inner Soddy circle of the three mutually tangent circles of radii $1, 2, 3$: a smaller fourth sphere cannot touch all three. When no face is right-angled, the Cayley-Menger determinant is the reliable route.

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2.25 Problem 25

[Statement]

Deal to Alan first, then Dora. Alan's three marbles must come from the 8 non-red ones; if he receives b blue and $3-b$ white, this has probability $\binom{4}{b}\binom{4}{3-b}/\binom{12}{3}$. Dora's three must then come from the $9-(4-b)=5+b$ remaining non-blue marbles, with probability $\binom{5+b}{3}/\binom{9}{3}$. Summing over $b=0,1,2,3$,

$$\frac{1}{220\cdot 84}\Big(4\cdot 10+24\cdot 20+24\cdot 35+4\cdot 56\Big)=\frac{1584}{18480}=\frac{3}{35}.$$

Answer. $\boxed{38}$

2.26 Problem 26

[Statement]

By AM-GM on four terms,

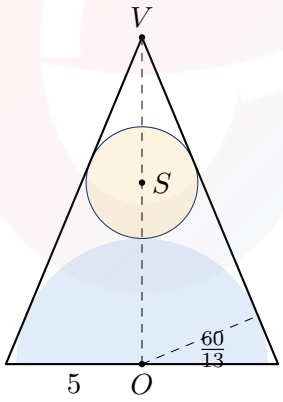
$$3x+y^3=x+x+x+y^3\ \geq\ 4\sqrt[4]{x^3y^3}=4\cdot 16^{3/4}=32,$$

with equality when $x=y^3$; together with $xy=16$ this gives $y=2$, $x=8$. So $m=32$ and $x+y+m=42$.

Answer. $\boxed{42}$

2.27 Problem 27

[Statement]



Work in an axial cross-section: an isosceles triangle with apex V, base of half-width 5 and height 12, so the slant side has length 13 and equation $12x+5y=60$. The hemisphere becomes a semicircle centred at the base centre O, and its radius is the distance from O to the slant side, $\frac{5\cdot 12}{13}=\frac{60}{13}$.

A sphere tangent to the lateral surface and to the hemisphere, and as large as possible, sits on the axis above the hemisphere. (Off-axis spheres in the two corner gaps near the base are tiny, and moving the axial sphere sideways only shrinks it.) Let its centre be $S=(0,h)$ with radius r. Tangency to the side gives $\dfrac{60-5h}{13}=r$; tangency to the hemisphere gives $h=r+\frac{60}{13}$. Substituting,

$$60-5r-\frac{300}{13}=13r\ \Longrightarrow\ 18r=\frac{480}{13}\ \Longrightarrow\ r=\frac{80}{39}.$$

Answer. $\boxed{119}$

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2.28 Problem 28

[Statement]

Rationalising, $\dfrac{2n^2}{1+\sqrt{1+4n^4}} = \dfrac{\sqrt{1+4n^4}-1}{2n^2}$, so

$$a_n = \frac{2n^2+1-\sqrt{1+4n^4}}{2n^2}.$$

Now use Sophie Germain: $1+4n^4 = (2n^2+2n+1)(2n^2-2n+1) = uv$ with $u+v = 2(2n^2+1)$. Then $2n^2+1-\sqrt{uv} = \frac{1}{2}(\sqrt{u}-\sqrt{v})^2$, hence

$$\sqrt{a_n} = \frac{\sqrt{u}-\sqrt{v}}{2n}, \qquad n\sqrt{a_n} = \frac{\sqrt{n^2+(n+1)^2}-\sqrt{(n-1)^2+n^2}}{2}.$$

The sum telescopes:

$$\sum_{n=1}^{119} n\sqrt{a_n} = \frac{\sqrt{119^2+120^2}-\sqrt{0^2+1^2}}{2} = \frac{\sqrt{28561}-1}{2} = \frac{169-1}{2} = 84.$$

Answer. $\boxed{84}$

Remark

The sum up to N is $\frac{1}{2}\left(\sqrt{N^2+(N+1)^2}-1\right)$, an integer precisely when $N^2+(N+1)^2$ is a perfect square, that is, when $(N, N+1, c)$ is a Pythagorean triple with consecutive legs. These N are $3, 20, 119, 696, \ldots$, generated by the Pell equation $(2N+1)^2 - 2c^2 = -1$. The problem's 119 is the third one, with $c = 169$.

Extension challenge

Show that $N_{k+1} = 6N_k - N_{k-1} + 2$ generates all such N, and find the next value after 696. (Answer: 4059, with $4059^2 + 4060^2 = 5741^2$.)

2.29 Problem 29

[Statement]

The rule forces each pile, read bottom to top, to be r_i red, then w_i white, then b_i blue bricks, with $r_i + w_i + b_i = 6$ and any of the three counts possibly zero. A pattern is therefore a pair of nonnegative triples (r_1, r_2, r_3), (b_1, b_2, b_3) with $\sum r_i = 6$, $\sum b_i = 6$ and $r_i + b_i \le 6$ for each i; the whites then fill in automatically, and their total is $18 - 12 = 6$ as required.

Without the constraint $r_i + b_i \le 6$ there are $\binom{8}{2}^2 = 784$ pairs. If pile i violates it, $r_i + b_i \ge 7$, and two piles cannot both violate it since $14 > 12$. For pile 1 with $r_1 = k \ge 1$ and $b_1 = j \in [7-k, 6]$, the other two piles can be completed in $(7-k)(7-j)$ ways, so the number of violations is

$$\sum_{k=1}^{6}(7-k)\sum_{t=1}^{k} t = 6\cdot 1 + 5\cdot 3 + 4\cdot 6 + 3\cdot 10 + 2\cdot 15 + 1\cdot 21 = 126.$$

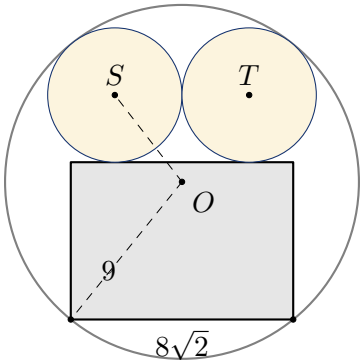
Three piles give $3 \cdot 126 = 378$ bad pairs, leaving $784 - 378 = 406$.

Answer. $\boxed{406}$

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2.30 Problem 30

[Statement]



Cross-section through a diagonal plane of the cube.

Put the centre O of the large sphere at the origin with the cube's axis vertical. The bottom face is a square of side 8, so its vertices are at horizontal distance $4\sqrt{2}$ from the axis; lying on the sphere, they satisfy $d^2 + 32 = 81$, so the bottom face is at height -7 and the top face at height $+1$.

A small sphere of radius r tangent to the top face has centre at height $1 + r$, and internal tangency to the large sphere puts the centre at distance $9 - r$ from O. Its horizontal distance from the axis is therefore

$$\rho = \sqrt{(9-r)^2 - (1+r)^2} = \sqrt{80 - 20r}.$$

Two such spheres can be tangent to each other only if $2r \le 2\rho$, that is $r^2 + 20r - 80 \le 0$, and the largest r is the root

$$r = -10 + \sqrt{180} = \sqrt{180} - 10.$$

At this value $\rho = r \approx 3.42 < 4$, so the contact points lie inside the top face, and the configuration is realised with the two centres diametrically opposite across the axis. Thus $m+n = 180+10 = 190$.

Answer. $\boxed{190}$

Remark

The bound $r \le \rho$ comes from the two centres lying on a horizontal circle of radius ρ; the tightest packing puts them at opposite ends of a diameter. With three congruent spheres arranged in an equilateral triangle the condition becomes $r \le \frac{\sqrt{3}}{2}\rho$, and with k spheres it is $r \le \rho \sin \frac{\pi}{k}$.

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3 Answer key

Problem	Answer	Problem	Answer	Problem	Answer
1	26	11	169	21	171
2	12	12	498	22	3480
3	24	13	136	23	112
4	38	14	213	24	869
5	33	15	70	25	38
6	364	16	288	26	42
7	44	17	18	27	119
8	72	18	202	28	84
9	79	19	97	29	406
10	7	20	40	30	190

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